

Answer **all** questions.

1. **Image 1.1** shows a region of lowland heath on the Pembrokeshire coast.

Image 1.1



Heathland is dominated by grasses and shrubs, such as heather and gorse, and supports a very diverse range of wildlife. Grazing by large herbivores, such as sheep and ponies, maintains this habitat. However, if the area is not grazed, it develops into woodland.

Since the 19th century, more than 75% of lowland heaths in the UK have been lost with a resulting loss of biodiversity.

(a) State what is meant by biodiversity. [1]

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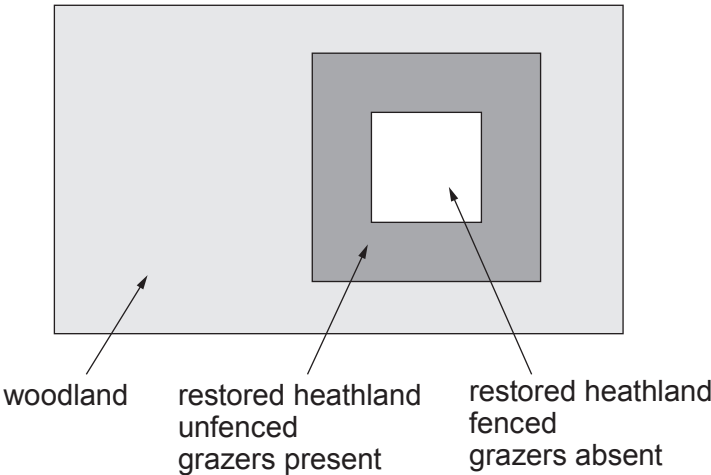
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In an investigation, an area of former heathland, which had developed into silver birch woodland, was cleared of trees to restore it to heathland. Within this area, a 50 m × 50 m square was fenced off to prevent the entry of grazing animals. Native ponies and sheep were allowed to graze the remaining area. This is shown in **Image 1.2**.

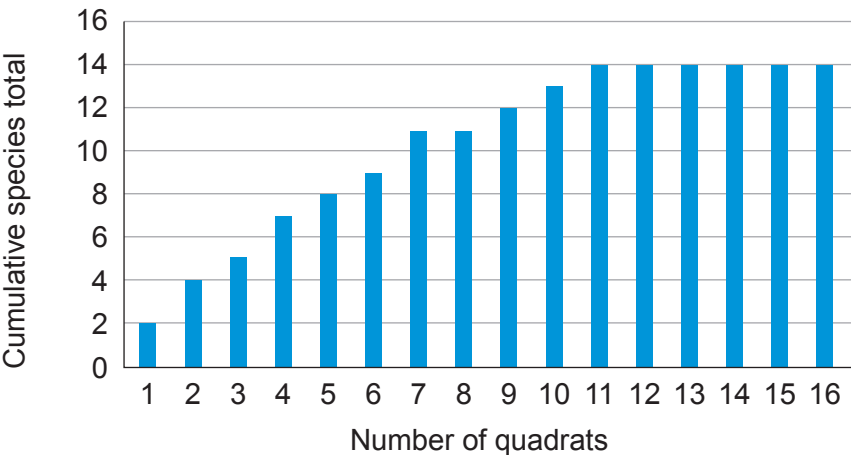
Image 1.2



The study area was left for **three** years, after which time a conservation group used Simpson's Diversity Index to assess the impact of grazing on the heathland biodiversity. They used 1 m² quadrats to sample both the restored heathland and the fenced area within it and recorded which species of plants were present.

Graph 1.3 shows the results of preliminary work to decide how many times to use the quadrat.

Graph 1.3



(b) It was decided to use 11 quadrats at each site. Explain the reasons for this decision. [2]

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- (c) **Table 1.4** shows the number of quadrats in which different species of plants were present.

Table 1.4

Common name	Scientific name	Relative height of plant	Number of quadrats in which the species was present	
			Fenced area	Unfenced area
Common bent grass	<i>Agrostis capillaris</i>	short	1	7
Sweet vernal grass	<i>Anthoxanthum odoratum</i>	short	1	4
Sedge	<i>Carex pilulifera</i>	short	1	3
Gorse	<i>Ulex europaeus</i>	medium	11	5
Bramble	<i>Rubus fruticosus</i>	medium	8	3
Silver birch	<i>Betula pendula</i>	tall	10	1
Ling heather	<i>Calluna vulgaris</i>	short	1	9
Bell heather	<i>Erica cinerea</i>	short	1	8

- (i) Name the genus to which bell heather belongs. [1]

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- (ii) Explain why the values in the table do not provide information about the actual number of organisms present. [1]

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- (iii) The unfenced area had a Simpson’s Diversity Index of 0.87.
The results for the fenced area are shown in **Table 1.5**.

Table 1.5

Common name	n	$(n - 1)$	$n(n - 1)$
Common bent grass	1		
Sweet vernal grass	1		
Sedge	1		
Gorse	11		
Bramble	8		
Silver birch	10		
Ling heather	1		
Bell heather	1		
$N =$		$\sum n(n - 1) =$	
$N(N - 1) =$			

Calculate the Simpson’s Diversity Index for this site.

[3]

$$D = 1 - \frac{\sum n(n - 1)}{N(N - 1)}$$

where;
 n = the number of quadrats where the species was present
 N = the total number of quadrats where species were found
 Σ = sum of

$D =$



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(d) Use all the information provided to answer the following questions.

(i) Conclude the effect of grazing on heathland biodiversity. [1]

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(ii) It was found that grazing increased the number of quadrats in which bell heather was found. Suggest an explanation for this. [2]

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